SEMINAR – RESEARCH

REAL-TIME SMOOTHING FILTER & 3D OBJECT TRACKING

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  - Results and analysis
- 3D Multi Object Tracking
  - Introduction
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Real-time Constrained Smoothing Filter



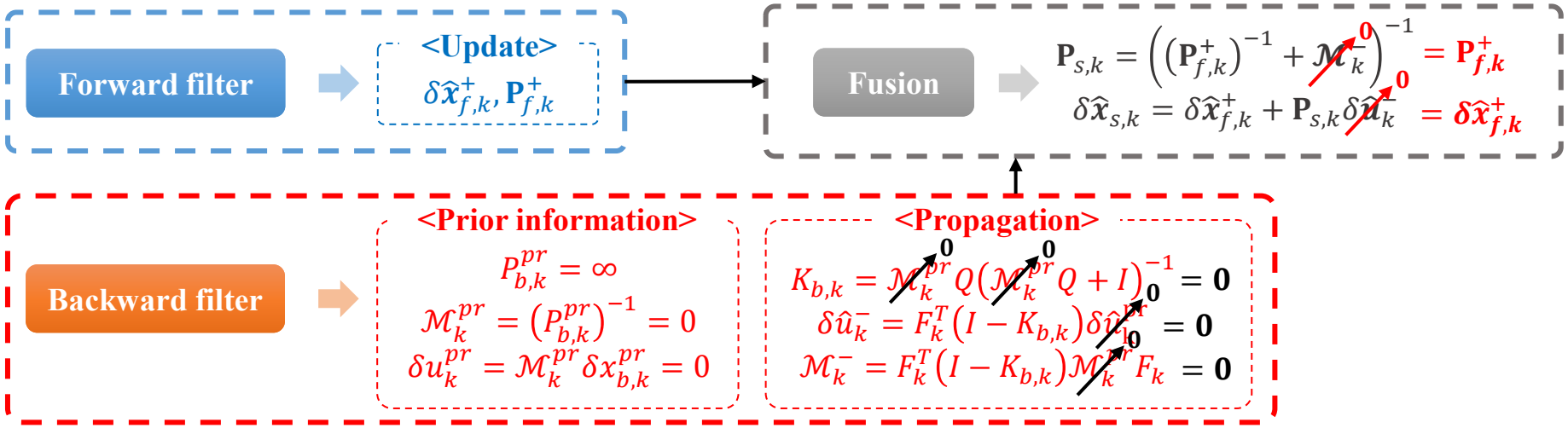
- Introduction
- Methodology
- Results and analysis

INTRODUCTION

- Research background
  - State estimation of system is important
  - Smoothing filter (SF)
    - Higher state estimation accuracy compared to a regular Kalman filter
    - **Limitation: Difficult to implement in real time**
  - Real-time smoothing filter (RSF)
    - Window formation method (via selection of appropriate window length)
    - **Limitation: Due to formal and structural limitations, there is no gain in the accuracy of current state estimation**
- Propose: **Real-time constrained smoothing filter (RCSF)**
  - Target: break the limitation of RSF in aspect of filter equation
  - Process
    - Define a prior information for the backward filter
    - Form constraints based on the prior information

LIMITATION OF THE RSF

- Performance of the real-time estimation
  - When we apply the RSF in current time $(i = k)$



  - This means that we cannot get compensation from the backward filter at the current time

- Cause of limitation: assumptions for prior information $\left(\mathbf{P}^{-}_{b,k} = \infty, \delta\hat{\boldsymbol{x}}^{-}_{b,k} = 0\right)$

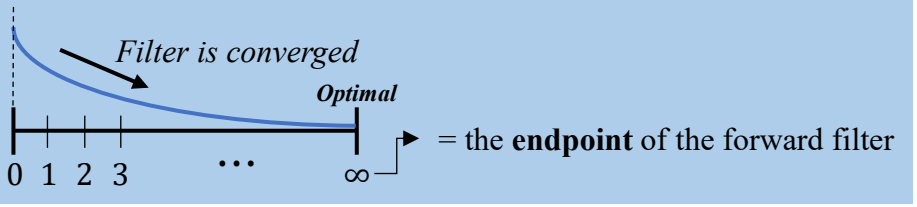
CONCEPT

Forward filter

- ✓ The forward filter is the same as the Kalman filter
- ✓ Kalman filter is **an optimal filter that minimizes the error covariance** of the system state information

< Theoretic method to **estimate the optimal minimum error covariance** >

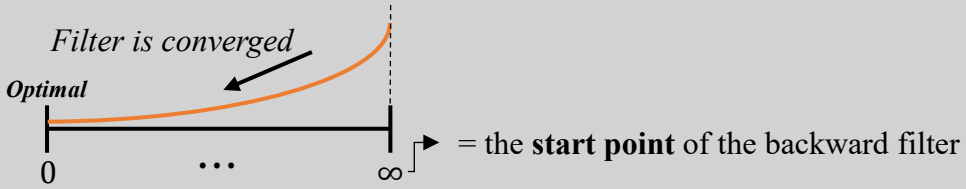
- We can get the optimal solution when the Kalman filter is converged until **infinity**



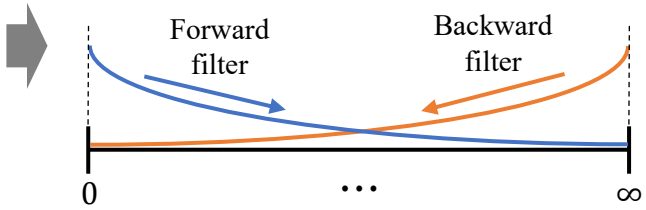
From this optimal perspective

Backward filter

- ✓ The backward filter is started from the endpoint of the forward filter



Execution range of the smoothing filter



DEFINITION OF THE PRIOR INFORMATION

- Calculate an optimal information matrix $\mathcal{M}_{opt}^{-} = \left(P_{b,opt}^{-}\right)^{-1}$
  - Forming a "discrete-time backward Riccati equation"

Discrete-time system and measurement model

$$\delta \boldsymbol{x}_{b,k-1} = \mathbf{F}_{k-1}^{-1}\delta \boldsymbol{x}_{b,k} + \mathbf{F}_{k-1}^{-1}\boldsymbol{\omega}_k$$
$$\delta \boldsymbol{y}_{k-1} = \mathbf{H}_{k-1}\delta \boldsymbol{x}_{b,k-1} + \boldsymbol{v}_{k-1}$$

Time Propagation

$$\mathbf{K}_{b,k-1} = \mathcal{M}_k^{+}\mathbf{Q}(\mathcal{M}_k^{+}\mathbf{Q} + \mathbf{I})^{-1}$$
$$\mathcal{M}_{k-1}^{-} = \mathbf{F}_{k-1}^{T}\left(\mathbf{I} - \mathbf{K}_{b,k-1}\right)\mathcal{M}_k^{+}\mathbf{F}_{k-1}$$

Measurement update

$$\mathcal{M}_{k-1}^{+} = \mathcal{M}_{k-1}^{-} + \mathbf{H}_{k-1}^{T}\mathbf{R}^{-1}\mathbf{H}_{k-1}$$

$$\begin{aligned}\mathcal{M}_{k-1}^{-} &= \mathbf{F}_{k-1}^{T}\mathcal{M}_k^{+}\mathbf{F}_{k-1} - \mathbf{F}_{k-1}^{T}\mathbf{K}_{b,k-1}\mathcal{M}_k^{+}\mathbf{F}_{k-1} \\ &= \mathbf{F}_{k-1}^{T}\mathcal{M}_k^{+}\mathbf{F}_{k-1} - \mathbf{F}_{k-1}^{T}\mathcal{M}_k^{+}\mathbf{Q}(\mathcal{M}_k^{+}\mathbf{Q} + \mathbf{I})^{-1}\mathcal{M}_k^{+}\mathbf{F}_{k-1}\end{aligned}$$

$$\begin{aligned}\mathcal{M}_{k-1}^{-} &= \mathbf{F}_{k-1}^{T}(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H})\mathbf{F}_{k-1} \\ &\quad -\mathbf{F}_{k-1}^{T}(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H})\mathbf{Q} \\ &\quad \{(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H})\mathbf{Q} + \mathbf{I}\}^{-1}(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H})\mathbf{F}_{k-1} \\ &= \mathbf{F}_{k-1}^{T}\mathcal{M}_k^{-}\mathbf{F}_{k-1} + \mathbf{F}_{k-1}^{T}\mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}\mathbf{F}_{k-1} \\ &\quad -\mathbf{F}_{k-1}^{T}\mathcal{M}_k^{-}\{(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}) + \mathbf{Q}^{-1}\}^{-1}\mathcal{M}_k^{-}\mathbf{F}_{k-1} \\ &\quad -\mathbf{F}_{k-1}^{T}\mathcal{M}_k^{-}\{(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}) + \mathbf{Q}^{-1}\}^{-1}\mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}\mathbf{F}_{k-1} \\ &\quad -\mathbf{F}_{k-1}^{T}\mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}\{(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}) + \mathbf{Q}^{-1}\}^{-1}\mathcal{M}_k^{-}\mathbf{F}_{k-1} \\ &\quad -\mathbf{F}_{k-1}^{T}\mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}\{(\mathcal{M}_k^{-} + \mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}) + \mathbf{Q}^{-1}\}^{-1}\mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}\mathbf{F}_{k-1}\end{aligned}$$

- If we solve the discrete-time backward Riccati equation by using any numerical method, we can get the optimal information matrix $\mathcal{M}_0^{-} = \mathcal{M}_{opt}^{-}$

CONSTRAINTS FOR THE BACKWARD FILTER

- Forming the constraints at the current time $i = k$
  - Define the prior information by taking a weighed average of the information of the forward filter and the optimal information of the backward filter

Information of the forward filter

✓ Immediately after the measurement outage area ends, both $\delta\hat{\boldsymbol{x}}_i^-$ and $\delta\boldsymbol{y}_i$ are inaccurate

✓ Thus, we use both information by taking the weighted average method

$$\mathbf{P}_{f,i,new} = \left(\left(\mathbf{P}_{f,i}^-\right)^{-1} + \left(\mathbf{P}_{f,i}^+\right)^{-1}\right)^{-1}$$

$$\delta\hat{\boldsymbol{x}}_{f,i,new} = \left(\mathbf{P}_{f,i}^-\right)^{-1}\left(\left(\mathbf{P}_{f,i}^-\right)^{-1} + \left(\mathbf{P}_{f,i}^+\right)^{-1}\right)^{-1}\delta\hat{\boldsymbol{x}}_{f,i}^- + \left(\mathbf{P}_{f,i}^+\right)^{-1}\left(\left(\mathbf{P}_{f,i}^-\right)^{-1} + \left(\mathbf{P}_{f,i}^+\right)^{-1}\right)^{-1}\delta\hat{\boldsymbol{x}}_{f,i}^+$$

Defining the prior information of the backward filter

✓ It is necessary to define the prior information that reflects information about the current time

✓ Thus, we define the prior information by taking a weighted average of $\mathbf{P}_{f,i,new}$ and $\boldsymbol{\mathcal{M}}_{opt}^-$

$$\begin{aligned}(\boldsymbol{\mathcal{H}}_i^-)^{-1} &= \mathbf{P}_{b,i,new}^- \\ &= \left(\left(\mathbf{P}_{f,i,new}\right)^{-1} + \boldsymbol{\mathcal{M}}_{opt}^-\right)^{-1}\end{aligned}$$

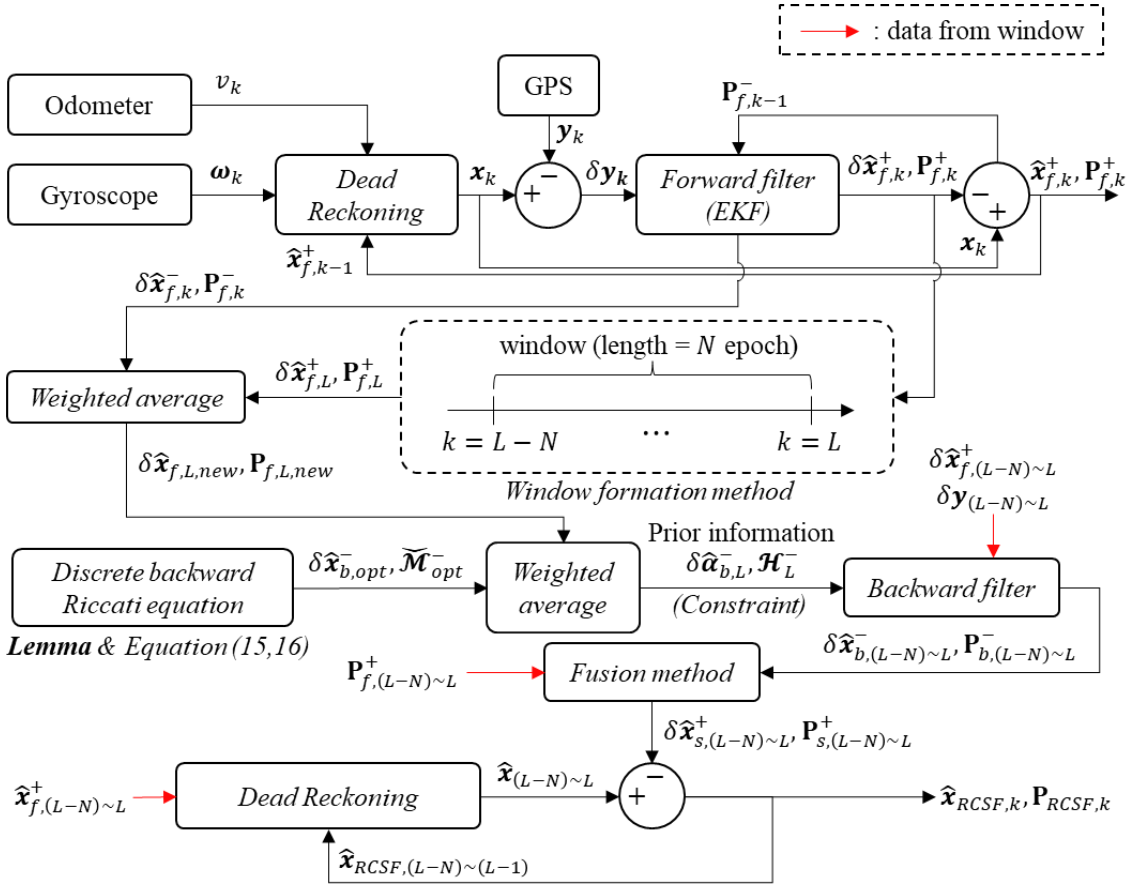
$$\begin{aligned}\delta\hat{\boldsymbol{x}}_{b,i,new}^- &= \left(\mathbf{P}_{f,i,new}\right)^{-1}\left(\boldsymbol{\mathcal{M}}_{opt}^- + \left(\mathbf{P}_{f,i,new}\right)^{-1}\right)^{-1}\delta\hat{\boldsymbol{x}}_{f,i,new} \\ &\quad + \boldsymbol{\mathcal{M}}_{opt}^-\left(\boldsymbol{\mathcal{M}}_{opt}^- + \left(\mathbf{P}_{f,i,new}\right)^{-1}\right)^{-1}\delta\hat{\boldsymbol{x}}_{b,opt}^- \\ &= \left(\mathbf{P}_{f,i,new}\right)^{-1}\left(\boldsymbol{\mathcal{M}}_{opt}^- + \left(\mathbf{P}_{f,i,new}\right)^{-1}\right)^{-1}\delta\hat{\boldsymbol{x}}_{f,i,new}\end{aligned}$$

Prior information matrix : $\boldsymbol{\mathcal{H}}_i^-$

Prior information state : $\delta\hat{\boldsymbol{\alpha}}_i^- = \boldsymbol{\mathcal{H}}_i^-\delta\hat{\boldsymbol{x}}_{b,i,new}^-$

DESIGN A NAVIGATION SYSTEM

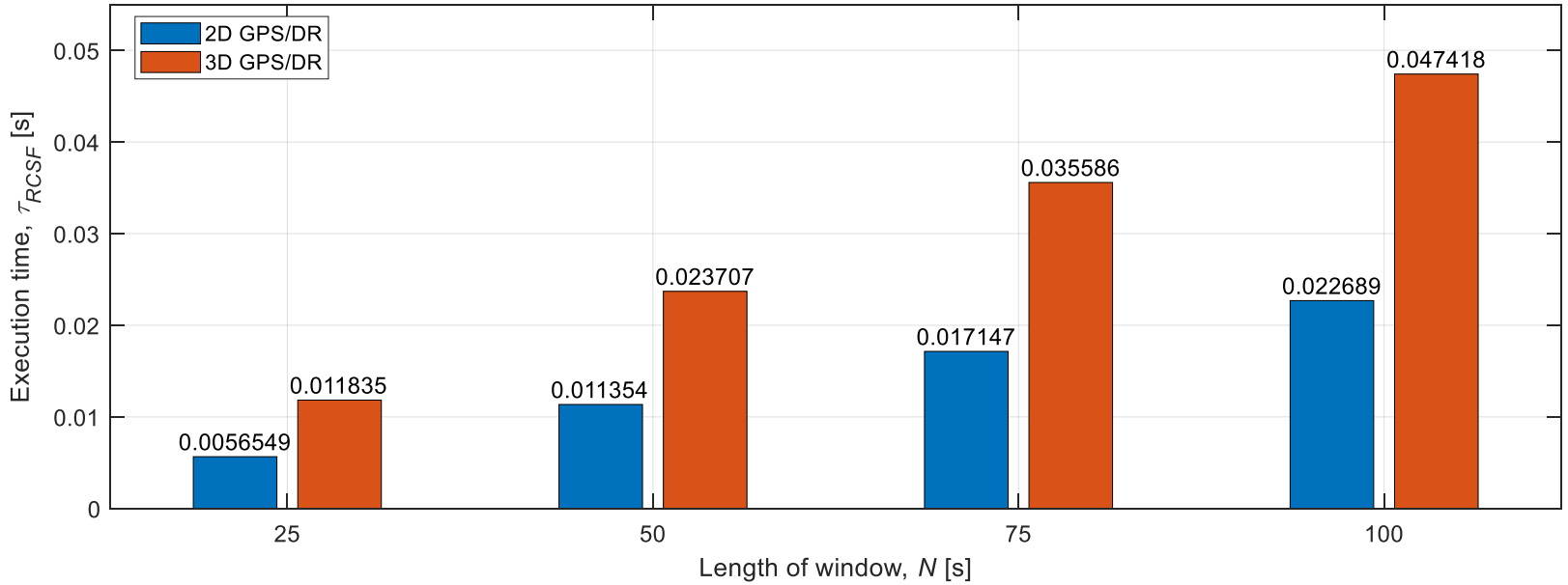
- To analyze the results of the RCSF, design a 3D GPS/DR navigation system



< Structure of RCSF-based navigation system >

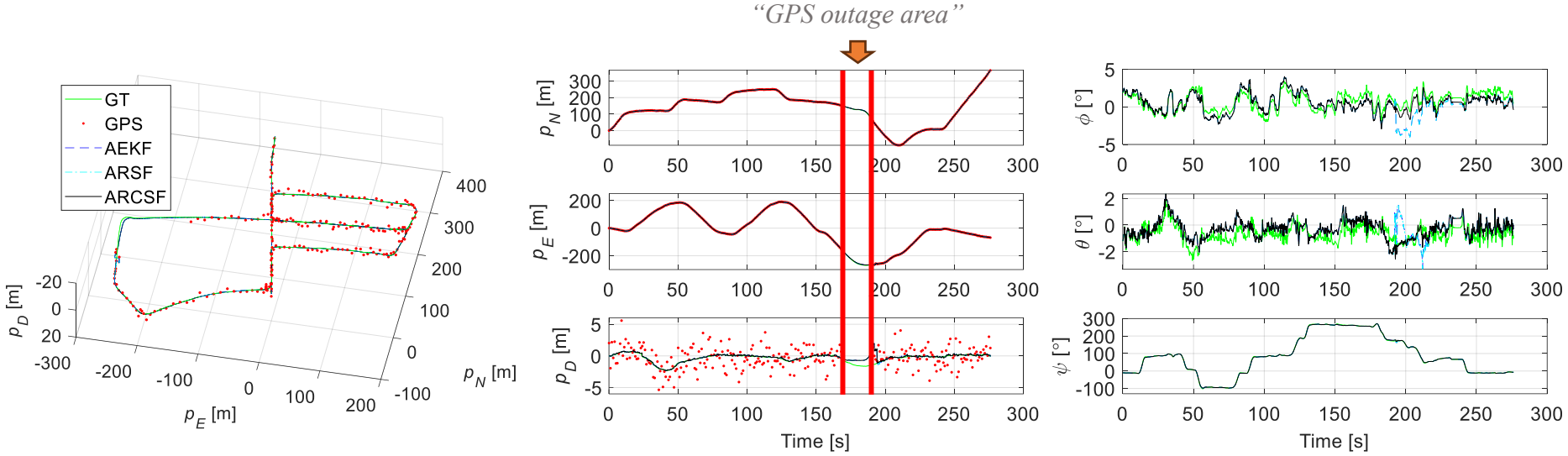
EXECUTION TIME ANALYSIS

- Assess practical feasibility of the RCSF-based navigation system
  - Execution time VS. Length of window : compare {N = 25, 50, 75, 100 s}
  - Execution time VS. System complexity : compare 2D(4th) with 3D(9th) system

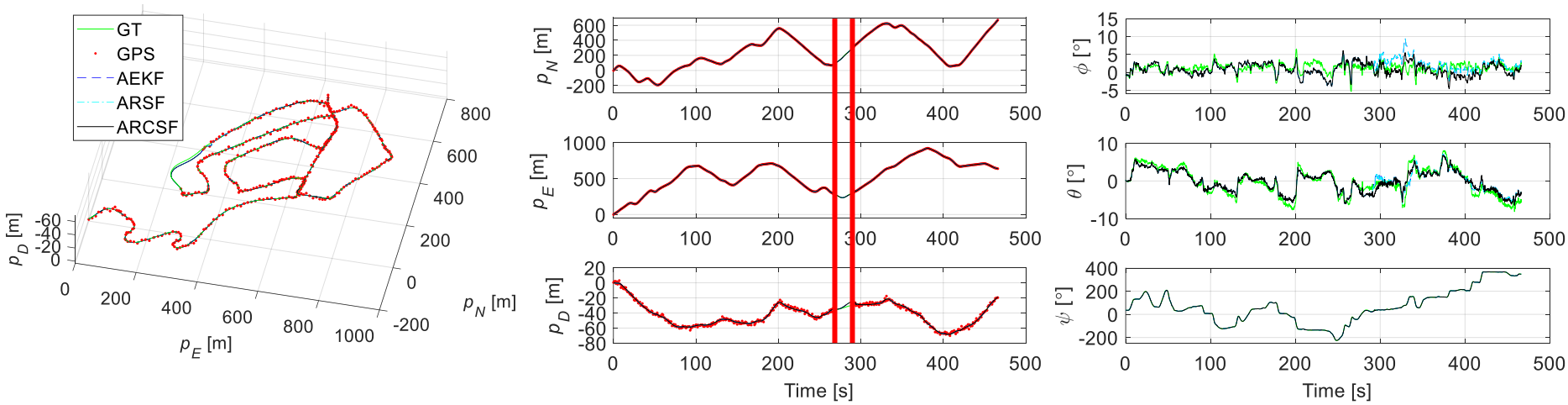


< Execution time according to the length of window and system complexity >

EXPERIMENTAL RESULTS

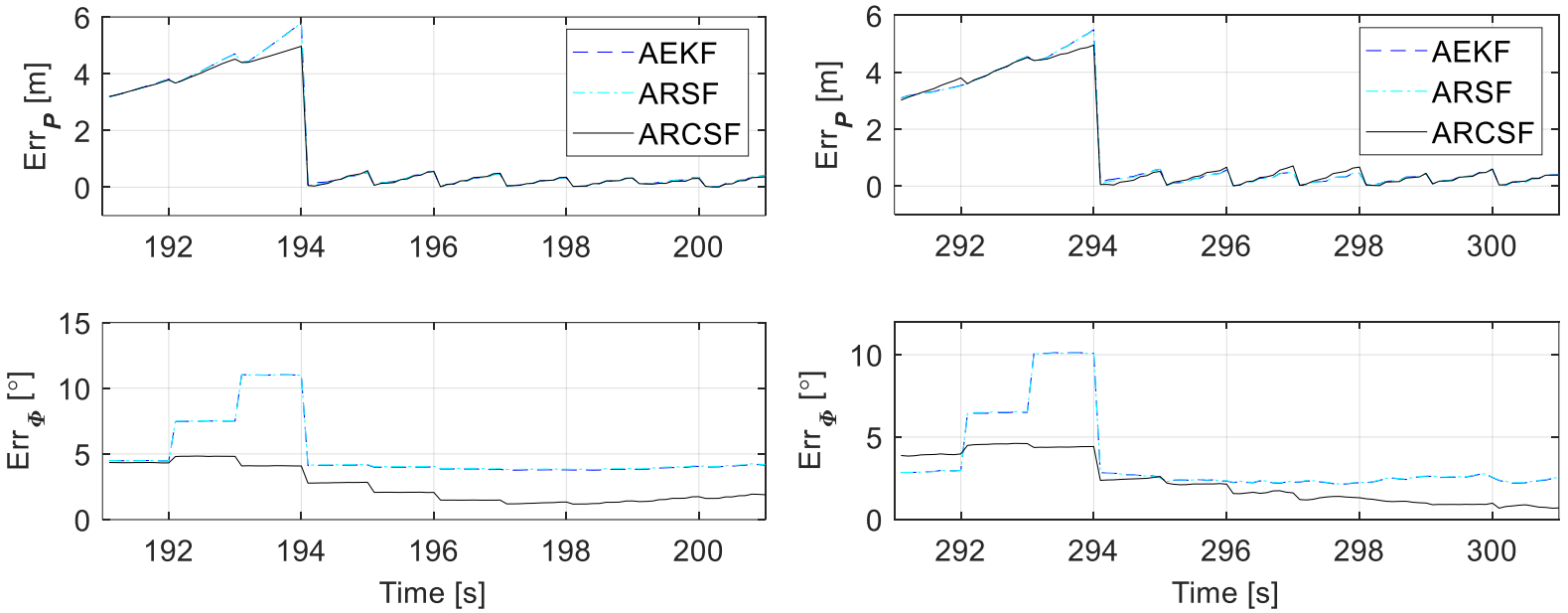


< Results of the RCSF-based 3D GPS/DR navigation (KITTI#1) >



< Results of the RCSF-based 3D GPS/DR navigation (KITTI#2) >

EXPERIMENTAL RESULTS



< Error in the real-time state estimation of the RCSF-based 3D GPS/DR navigation (Left: #1, Right: #2) >

< Position and attitude RMSE of the 3D GPS/DR navigation system >

Applied Filter	RMSE (KITTI #1)		RMSE (KITTI #2)	
	Position [m]	Attitude [°]	Position [m]	Attitude [°]
AEKF	2.3509	5.5221	2.2795	4.3850
ARSF	2.3503	5.5455	2.2774	4.3914
ARCSF	**2.2503**	**2.8550**	**2.2439**	**2.6936**

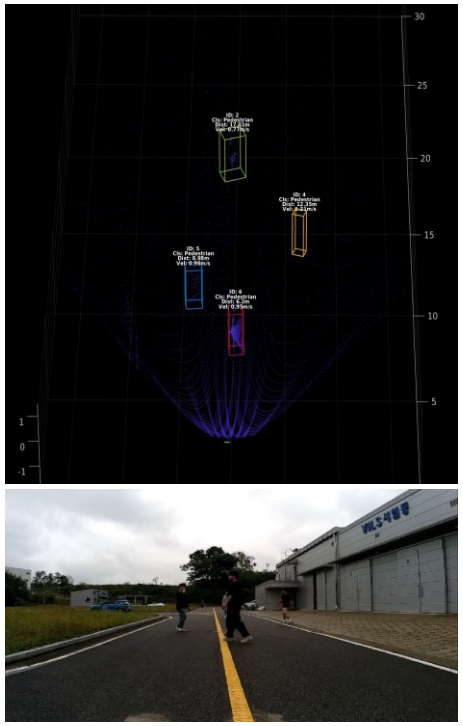
3D Multi Object Tracking



- Introduction
- Environments
- Research process
- Future works

INTRODUCTION

- Research background
  - Sensor project (UVARC)
    - Target 1: Sensor fusion algorithm – LiDAR, RADAR, and Stereo camera
    - Target 2: Detection and Recognition + Tracking



< Result of detection and recognition[1] >

▶ Limitation

✓ Detection results (bounding box) are discontinuous

- Two or more objects overlap
- Insufficient data

▶ Research target

✓ Intermediate fusion

- (LiDAR or RADAR) with camera sensor

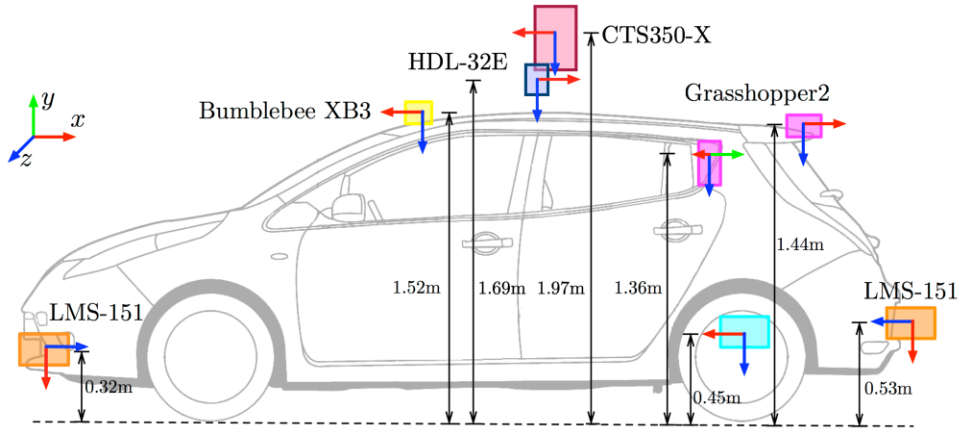
✓ Filter-based Seamless 3D multi-object tracking

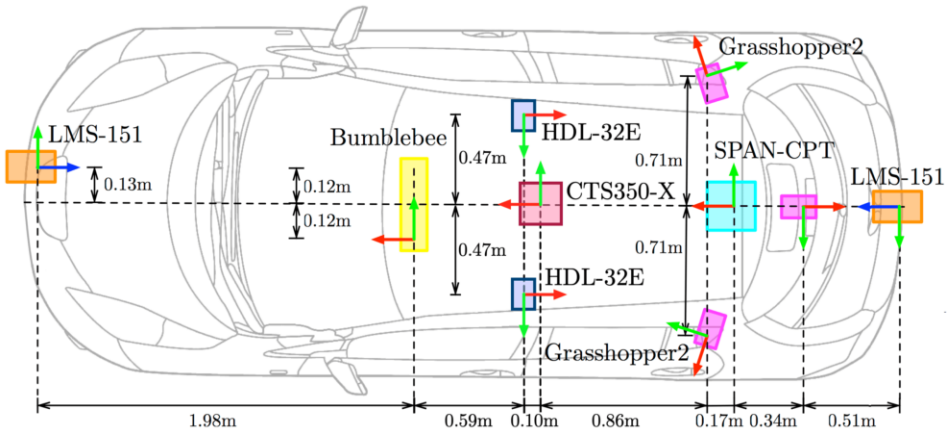
- Interacting multiple model (IMM) filter
- Particle filter
- Random finite set filter

[1] *Sensor Project (UVARC)*, 2024

ENVIRONMENTS

- Oxford Radar Robotcar dataset





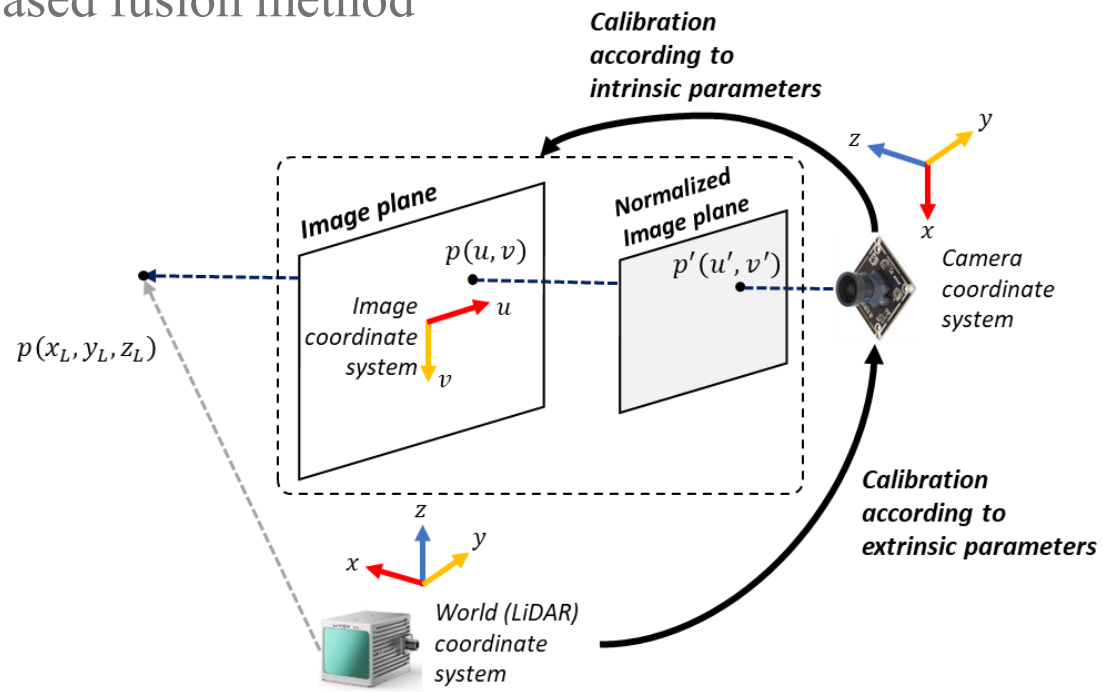
Stereo camera
Point Grey Bumblebee XB3 trinocular stereo camera
1280x960x3
16Hz
Global shutter / 3.8mm lens
66°(HFoV), 12cm baseline

3D LiDAR
Velodyne HDL-32E 3D LIDAR
360°(HFoV)x41.3°(VFoV)
20Hz
32 planes
100m range (max.)
2cm range resolution

2D RADAR
Navtech CTS350-X Millimetre-Wave FMCW radar
4Hz
400 measurements per rotation
163m range (max.)
4.38cm range resolution
1.8° beamwidth

EARLY FUSION PROCESS

- Calibration-based fusion method

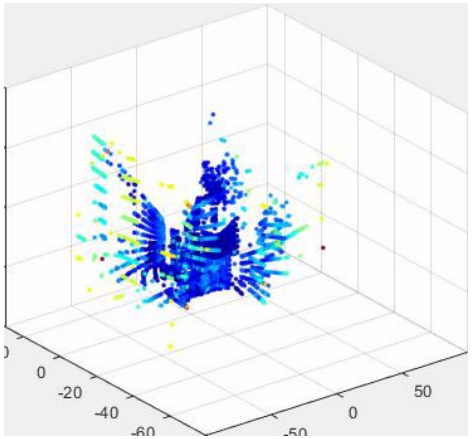


< Process of calibration-based fusion >

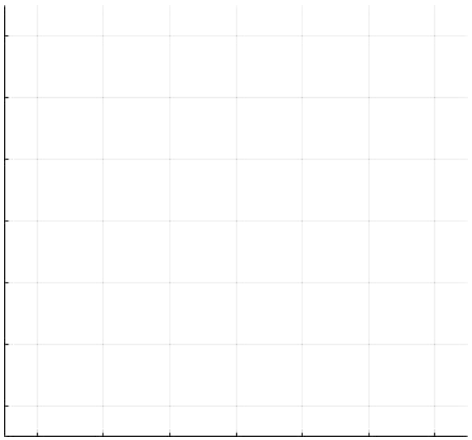
Formula about {point-cloud to image} calibration

$$\boldsymbol{p}'(u',v') = \begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \end{bmatrix} \begin{bmatrix} \mathbf{I}_{3\times3} \\ \mathbf{0}_{1\times3} \end{bmatrix}^T \begin{bmatrix} \mathbf{R}_L^C & \boldsymbol{T}_L^C \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} x_L \\ y_L \\ z_L \\ 1 \end{bmatrix} \qquad \boldsymbol{p}(u,v) = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{u'}{w'} & \frac{v'}{w'} & 1 \end{bmatrix}^T$$

EARLY FUSION RESULTS











SIMPLE DETECTION RESULT – YOLO









FUTURE WORKS

- Investigate prior studies
  - AI-based 3D multi-object tracking
  - Model-based 3D multi-object tracking
  - Evaluation technique of tracking – MOTA, MOTP, etc..

- Next studies
  - YOLO method-based detection → Bounding box
  - Intermediate fusion – YOLO + LiDAR & Radar raw
  - Filter-based multi-object tracking
    - Interacting multiple model (IMM) filter
    - Particle filter
    - Etc.

Q & A

